

PROJECT AND TEAM INFORMATION

Project Title

Campus WiFi Mesh Topology Optimizer

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Modern educational institutions, such as Graphic Era University, rely heavily on robust wireless infrastructure to support academic services, digital learning platforms, research labs, and administrative operations. Providing seamless wireless connectivity across multi-building university campuses requires deploying numerous Access Points (APs). However, establishing redundant inter-AP links creates severe network over-meshing, increases wireless co-channel interference, elevates multi-hop latency, and needlessly increases hardware deployment and operational costs.

Unplanned mesh networks frequently suffer from signal collisions and packet drops when APs connect indiscriminately to nearby nodes. Campus network managers lack lightweight, algorithmic simulation tools to evaluate and design optimized wireless mesh topologies before physical hardware installation. Addressing this challenge is crucial to ensuring cost-effective, high-performance, and low-latency campus connectivity.

State of the Art / Current solution

Currently, campus WiFi mesh networks are designed manually using trial-and-error physical site surveys or high-cost, proprietary enterprise site-survey software. Traditional approaches often rely on ad-hoc AP placement, leading to indiscriminate node interconnectivity. This over-meshing produces co-channel congestion and multi-hop latency delays because data packets traverse unnecessary intermediate APs before reaching the primary campus internet gateway. Moreover, existing commercial tools are often complex, black-boxed, expensive, and lack custom algorithmic flexibility to optimize network topologies based on combined multi-variable metrics like user density, signal interference, and structural layout.

Project Goals and Milestones

The primary goal of the 'Campus WiFi Mesh Topology Optimizer' is to construct an algorithmically driven, software-based simulation framework that models campus layouts, evaluates inter-AP wireless links, and derives an optimized mesh network topology balancing coverage, performance, and infrastructure costs.

Key Milestones:

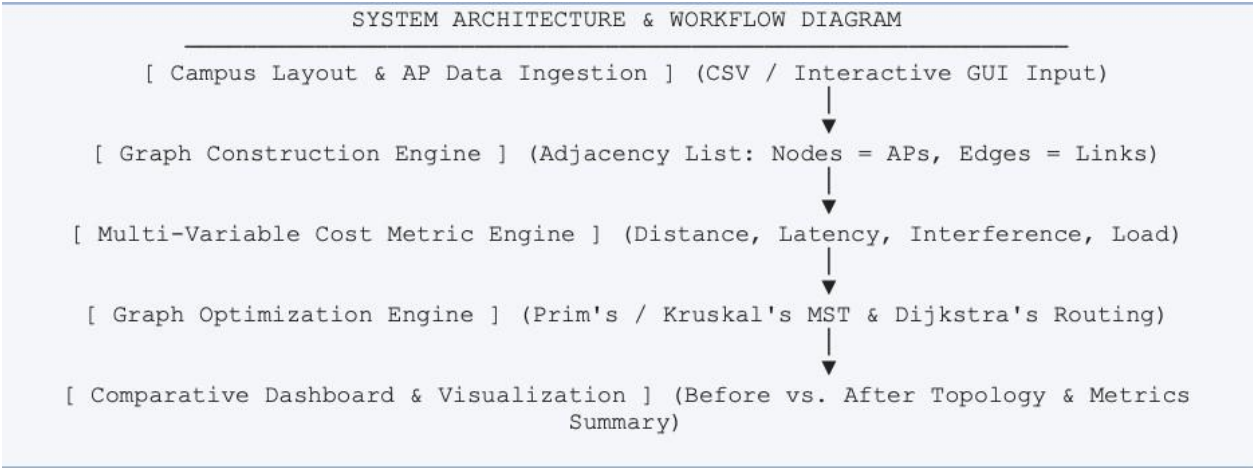
- Milestone 1: Data Ingestion & Graph Abstraction Module — Model campus APs as vertices and potential wireless links as weighted edges.
- Milestone 2: Multi-Variable Cost Engine — Construct a comprehensive link cost function combining physical distance, latency, signal strength, interference, and user load.
- Milestone 3: Algorithmic Optimization Module — Implement Prim's and Kruskal's MST algorithms for edge pruning, and Dijkstra's algorithm for shortest-path routing to the primary gateway.
- Milestone 4: Interactive Visual Dashboard — Build a user interface (Streamlit/Tkinter/NetworkX) displaying 'Before vs. After' network graph visualizations and comparative performance metrics.

Project Approach

The proposed solution follows a modular software architecture separating data ingestion, graph construction, multi-metric cost calculation, algorithmic optimization, and visual output rendering.

1. Graph Representation: The campus layout is modeled as a weighted graph $G = (V, E)$ using Adjacency Lists. Access Points represent vertices (V) and wireless links represent candidate edges (E).
2. Multi-Metric Cost Model: Edge weights are calculated using a custom cost function incorporating physical distance, estimated latency, radio frequency interference (2.4GHz / 5GHz), and simulated user load.
3. Core Algorithms: Prim's and Kruskal's Minimum Spanning Tree (MST) algorithms prune redundant, high-cost links while ensuring 100% graph connectivity. Dijkstra's Algorithm determines the lowest-latency routing path from edge nodes to the central gateway.
4. Technology Stack: Developed in Python 3.x using NetworkX and Matplotlib for graph rendering, Pandas/NumPy for dataset parsing, and Streamlit/Tkinter for the interactive administrative GUI.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

- Software Simulation Application: A Python-based desktop/web dashboard allowing administrators to input campus AP coordinates and wireless parameters.
- Optimized Mesh Topologies: Visual graph maps rendering 'Before Optimization' (dense, over-meshed) versus 'After Optimization' (pruned MST) network structures.
- Quantitative Metric Evaluation Report: Automated performance summary showing active link counts, average latency (ms), overall interference reduction, and total topology cost index.
- Algorithmic Implementation Codebase: Fully documented DSA implementation featuring custom graph structures, Prim's, Kruskal's, and Dijkstra's algorithms.

Assumptions

- Simulation Environment: The system operates as a decision-support simulation tool and does not reconfigure physical campus hardware directly.
- Static Physical Layout: Input parameters (AP coordinates, building obstacles, baseline RF propagation) are static during individual simulation runs.
- AP Line-of-Sight & Compatibility: All candidate inter-AP links are assumed to operate within standard 802.11s mesh specs and compatible frequency bands.

References

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- Prim, R. C. (1957). Shortest connection networks and some generalizations. Bell System Technical Journal, 36(6), 1389-1401.
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